Course: Computer Vision

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Slot: D

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Submitted By

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Hough Circle Transform

The Hough Circle Transform is used to detect circles in an image. For an edge point and a possible circle center, we calculate the radius using the distance between them.

Where:

- (x,y) = coordinates of the edge point
- (a,b) = coordinates of the candidate center
- r = radius of the circle

1. Calculate the radius for each combination

Given

Edge points:

P1 =(5,4) P2 =(7,6)

Candidate centers:

C1 =(2,4) C2 =(4,3)

We need to calculate the radius for all four combinations:

1. $P1 - C1$
2. $P1 - C2$
3. $P2 - C1$
4. $P2 - C2$

Combination 1: P_1 and C_1

$P1 = (5,4), C1 = (2,4)$

Using the distance formula:

$r=(5-2)2+(4-4)2$

Calculate each difference:

$5-2=3$ $4-4=0$

Therefore:

$r=32+02$ $=9$ $r=3$

Combination 2: P_1 and C_2

$P1 = (5,4), C2 = (4,3)$ $r=(5-4)2+(4-3)2$ $=12+12$ $=1+1$ $=2$ $r \approx 1.41$

Combination 3: P_2 and C_1

$P2 = (7,6), C1 = (2,4)$ $r=(7-2)2+(6-4)2$ $=52+22$ $=25+4$ $=29$ $r \approx 5.39$

Combination 4: P_2 and C_2

$P2 = (7,6), C2 = (4,3)$ $r=(7-4)2+(6-3)2$ $=32+32$ $=9+9$ $=18$ $r \approx 4.24$

Results Table

Edge Point	Candidate Center	Radius
P_1 (5,4)	C_1 (2,4)	3.00
P_1 (5,4)	C_2 (4,3)	1.41
P_2 (7,6)	C_1 (2,4)	5.39
P_2 (7,6)	C_2 (4,3)	4.24

Identifying the matching radius

For a particular center to represent the correct circle, different edge points should produce approximately the same radius.

For C_1:

- $P_1 \rightarrow 3.00$
- $P_2 \rightarrow 5.39$

They are not equal.

For **C_2**:

- $P_1 \rightarrow 1.41$
- $P_2 \rightarrow 4.24$

They are also not equal.

Neither C_1 nor C_2 gives matching radius values for the two given edge points.

Therefore, neither candidate center can be identified as the correct circle center based on these two points alone.

2. Calculate the radius

Given

Edge point:

$P=(8,6)$

Candidate center:

$C=(4,3)$

Use:

$r=(x-a)2+(y-b)2$

Substitute the values:

$r=(8-4)2+(6-3)2$

Calculate:

$8-4=4$ $6-3=3$

Therefore:

$r=42+32$ $=16+9$ $=25$ $r=5$

Interpretation

The distance between the candidate center (4,3) and the edge point (8,6) is 5 pixels/units. Therefore, if this point belongs to the detected circle, the circle has a radius of 5.

3. Calculate the radius

Given

Center:

$C=(5,4)$

Edge point:

$P=(8,8)$

Using the distance formula:

$r=(x-a)2+(y-b)2$

Substitute:

$r=(8-5)2+(8-4)2$

Calculate:

$8-5=3$ $8-4=4$

Therefore:

$r=32+42$ $=9+16$ $=25$ $r=5$

Interpretation

The edge point (8,8) is exactly 5 units away from the center (5,4). Hence, the detected circle has a radius of 5 units.